



RIVER VALLEY HIGH SCHOOL

JC 2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

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CENTRE
NUMBER

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INDEX
NUMBER

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H3 BIOLOGY

9816/01

Paper 1

24 September 2021

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the space provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiners' Use	
Section A	
1	
2	
Section B	
3 / 4	
Total	

Section A

Answer **all** the questions in this section.

- 1** You should read through the whole of the insert and **Question 1** carefully and then use the information provided to answer the questions. The insert includes Table 1.1, Fig. 1.1, Fig. 1.2, Fig. 1.3, Fig. 1.4 and Fig. 1.5.

The entire question is based on information about evolution of multicellularity in animals.

- (a)** Discuss the extent to which the endosymbiotic origin of eukaryotes conforms to cell theory. [2]

- (b)** Using your knowledge on the central dogma and the findings on the mode of genetic transfer given in the insert, suggest how genetic transfer of mtDNA took place from the mitochondria to the nucleus. [3]

- (c) The mtDNA of extant species includes at least one of the genes coding for rRNA. Explain the reason for retaining at least one rRNA gene in the mtDNA. [2]

- (d) (i) Using Fig. 1.1 and Fig 1.2, identify three features and the related molecules that must have evolved **and** explain the significance of **each feature** in the formation of multicellular organisms. [3]

feature	molecules that must have evolved	significance in the formation of multicellular organisms
gene regulation		

- (ii) Unicellular organisms exist even after multicellular organisms evolved.
Describe one selective advantage of each type of organism. [2]

- (e) Using your knowledge of stem cells and Table 1.1, Fig 1.3 and Fig. 1.4, discuss how genetic and molecular differences between LUA and LCMA resulted in the differences observed in the multicellular stage. [5]

- (f) Compare the regulation of enzymes in a cell and the regulation of cells in a multicellular living organism. [4]

- (g) Studying how mutations in mtDNA accumulate with age is critical because of their association with numerous genetic diseases.

Using the information provided in the insert and Fig. 1.5, suggest and explain **two** ways in which knowledge of mtDNA mutations could be used combat genetic diseases in humans. [4]

[Total: 25]

- 2 Recent studies show that epigenetic diversity is involved in adaptation and survival of species in changing environments.

Compare epigenetic changes with mutations and discuss how epigenetic changes complement genetic diversity to contribute to long-term survival of species.

[25]

[illegible]

Section B

Answer **one** question in this section.

Either

- 3** The ongoing Covid-19 pandemic has resulted in worldwide disruption and millions of fatalities.

Describe the underlying causes of pandemics and discuss the difficulties in preventing future pandemics.

[25]

Or

- 4 The most important biomolecule of a cell is the phospholipid. Discuss.

[25]

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